COURSE REPORT: TEXT CLASSIFICATION ON NYT

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ABSTRACT

This course report completes three text-classification tasks on a shared NYT test set. Task 1 compares binary bag of words, Word Frequency and TF-IDF; Task 2 compares three sources of 100-dimensional mean-pooled word embeddings. Both tasks use logistic regression. Task 3 fine-tunes BERT-base-uncased with the required 64-token input and three training epochs. The cleaned corpus contains 13,007 documents in 29 categories, shuffled into an 80/10/10 split. All methods are evaluated using Accuracy and Macro-F1. In the required experiments, TF-IDF achieves the highest accuracy, 94.01%, and Word Frequency the highest Macro-F1, 83.61%. NYT Word2Vec leads the averaged embeddings at 81.85% Macro-F1, while prescribed-setting BERT reaches 62.97%. Supplementary controls vary input length, class weighting, pooling, training budget and text position, bringing the study to 233 configurations. Five partition seeds and three training seeds show how representation rankings depend on text access, class distribution and optimization. On the reference partition, ten-epoch uniform-loss BERT-512 reaches $86.38 \pm 0.51\%$ Macro-F1 and reverses the three-epoch loss ordering.

1 COURSEWORK OBJECTIVES AND REPORT ORGANIZATION

The coursework studies the principles, implementation and classification performance of different text representations. The report follows its three tasks: sparse lexical features with logistic regression in Task 1 (Section 3), averaged GloVe and Word2Vec embeddings in Task 2 (Section 4), and prescribed-setting BERT fine-tuning in Task 3 (Section 5). Each task presents its requirements, method, implementation settings and test results together. Table 6 collects all seven methods.

These representations preserve different information. Bag of words retains specific lexical cues, averaged embeddings compress the document into 100 semantic features, and BERT models contextual relationships within its visible tokens. The NYT corpus contains frequent sports topics and smaller policy and science categories, motivating both document-level accuracy and equally weighted Macro-F1. Section 6 adds controlled exploration of input access, class weighting, pooling and optimization budget to help explain differences in the required experiments.

2 SHARED DATA AND EXPERIMENTAL SETUP

2.1 DATA SOURCES AND CLEANING

The NYT data come from the public WeSHClass release associated with weakly supervised hierarchical text classification (Meng et al., 2019). The release's `dataset.txt` and `labels.txt` are aligned by line and converted to a table with `text` and `label` columns. The original labels are preserved as 29 classes. AG News is obtained from the public dataset distribution associated with Zhang et al. (2015); only its training-text column is used to learn word vectors. Resource URLs are provided in Appendix B.

The raw NYT release contains 13,081 records. Before splitting, two identical-text groups with conflicting labels are removed, eliminating four records. Exact-text duplicates in the remaining data are reduced to one record per document, removing another 70 records. The resulting corpus has 13,007 unique documents. No identical text is shared between partitions. Mean document length under the traditional tokenizer is 653.2 words, with a median of 686.

2.2 Common Split and Fitting Scope

A NumPy `RandomState(42)` permutation shuffles the cleaned row indices. The first $\lfloor 0.8N \rfloor$ indices form the training set, the next $\lfloor 0.1N \rfloor$ form the validation set, and the remaining indices form the test set. The resulting sizes are 10,405, 1,300, and 1,302. This is a shuffled split without stratification. Every main-comparison model uses these saved row indices.

The NYT vocabulary, TF-IDF weights, NYT Word2Vec model, standardization parameters, and classifier parameters are fitted on training data. The GloVe parameters and the pretrained BERT parameters are external resources; BERT is then fine-tuned using only the NYT training partition. The AG News Word2Vec model uses the external training corpus. Validation and test text are transformed after fitting the corresponding representation.

Validation data select each model configuration. Selected models are evaluated on the same 1,302 test documents, without refitting on validation data.

2.3 Evaluation Metrics and Class Distribution

Let M be the number of evaluation documents and $\hat{y}_d$ their predicted labels. Accuracy and Macro-F1 are computed as

$$\text{Accuracy} = \frac{1}{M}\sum_{d=1}^{M} \mathbb{1}[\hat{y}_d = y_d], \qquad \text{Macro-F1} = \frac{1}{29}\sum_{k=1}^{29} \frac{2\text{TP}_k}{2\text{TP}_k + \text{FP}_k + \text{FN}_k}. \tag{1}$$

Per-class F1 is defined as zero when its denominator is zero. Scores are reported as percentages, and differences between scores are stated in percentage points. Differences are computed from unrounded scores.

In the reference partition, training counts range from 22 documents for abortion to 1,795 for baseball. The seven sports categories account for 68.28% of the test set. Always predicting baseball, the most frequent training category, would give 16.28% test accuracy and 0.97% Macro-F1. The complete split counts in Appendix A show why an aggregate accuracy score alone gives limited information about performance across topics.

The 12 low-frequency classes contain only 82 of the 1,302 reference-test documents (6.30%), but constitute 41.38% of the equally weighted class terms in Macro-F1. A method can therefore improve accuracy by handling frequent topics better while losing recognition of several small topics. The two metrics answer different questions: how many documents are classified correctly, and how well topics are recognized when each class counts equally. Their disagreement is meaningful evidence about the task, rather than a contradiction between evaluation measures.

2.4 Implementation and Saved Records

The main comparison uses seed 42 for Python, NumPy, PyTorch, and the shuffled training loader. Word2Vec uses one worker, and the main CPU thread setting is eight. Repeated-partition classical runs use four CPU threads per worker; independent BERT configurations run on four H100 GPUs. The environment includes Python 3.10.16, scikit-learn 1.7.2, Gensim 4.4.0, PyTorch 2.9.0 with CUDA 12.6, and Transformers 4.57.3. Saved records include validation trials, split indices, predictions, class metrics, confusion matrices, and BERT training history. Appendix B gives execution commands and elapsed times.

3 Task 1: Bag of Words and Logistic Regression

Task 1 constructs binary bag of words, Word Frequency and TF-IDF representations, trains a logistic-regression classifier for each, and reports NYT test Accuracy and Macro-F1. The three methods share tokenization, the training vocabulary and classifier search, isolating differences between word presence, repeated counts and rarity weighting.

3.1 Tokenization and Sparse Lexical Features

For the six traditional models, text is lowercased and tokenized with the regular expression `[a-z]+(?:'[a-z]+)?`. This retains English alphabetic words and an internal apostrophe, preserves repeated occurrences, and excludes digits and punctuation. Stop words are retained. The vocabulary V contains every token observed in the NYT training partition, giving $|V| = 80,350$ on the reference split. Validation and test documents are transformed with this fixed vocabulary.

Let c_{di} be the number of occurrences of vocabulary word i in document d. Binary bag of words and Word Frequency use the same coordinates but assign different values:

$$x_{di}^{\text{binary}} = \mathbb{1}[c_{di} > 0], \qquad x_{di}^{\text{count}} = c_{di}. \tag{2}$$

Binary features record presence, while count features also retain repeated evidence. Neither representation records word order. A token absent from the training vocabulary contributes no feature at evaluation time.

TF-IDF weights the count of a word by its rarity across training documents. With N training documents and document frequency df_i, the implementation uses smoothed inverse document frequency and document-level normalization:

$$a_{di} = c_{di}\left[\log\left(\frac{1+N}{1+\mathrm{df}_i}\right)+1\right], \qquad x_d^{\text{tfidf}} = \frac{a_d}{\|a_d\|_2}. \tag{3}$$

The inverse document frequencies are fitted on training documents. The count term uses raw frequency, without logarithmic scaling. A document with no vocabulary words receives a zero vector. L2 normalization reduces the direct effect of document length on the scale of the features.

3.2 Logistic Regression and Model Selection

The six fixed representations use multinomial logistic regression. For a feature vector x and class k, its probability is

$$p(y = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_{j=1}^{29}\exp(w_j^\top x + b_j)}. \tag{4}$$

Prediction takes the class with the largest probability. Training minimizes multiclass cross-entropy with L2 regularization using the `lbfgs` solver and a maximum of 2,000 iterations. The inverse regularization parameter C is chosen separately for each representation from $\{0.1, 1, 10\}$.

For each representation, the highest validation Macro-F1 selects C, with ties resolved in favor of the smaller value. Binary, count and normalized TF-IDF values enter the classifier directly. The six traditional methods use scikit-learn (Pedregosa et al., 2011) and uniform example weights. Task 2 uses the same logistic-regression and selection rules.

3.3 Results and Analysis

Table 1 reports the Task 1 comparison. TF-IDF has the highest test accuracy, 94.01%, while Word Frequency has the highest test Macro-F1, 83.61%. Word Frequency also exceeds binary features on both metrics. Retaining repeated lexical evidence is useful in this run, and TF-IDF's normalization and rarity weighting produce the largest number of correct predictions within the main settings.

Table 1: Task 1 validation and test results (%). All three representations use logistic regression; validation Macro-F1 selects C.

	Validation		Test		
Method	Acc.	Macro-F1	Acc.	Macro-F1	Selection
Binary BoW	94.38	84.86	92.47	80.47	$C = 10$
Word Frequency	93.69	84.02	93.63	83.61	$C = 1$
TF-IDF	95.00	84.86	94.01	82.85	$C = 10$

The two leading sparse methods emphasize different aspects of performance. TF-IDF correctly classifies 1,224 of 1,302 test documents, while Word Frequency correctly classifies 1,219. Word

Frequency's higher Macro-F1 reflects its advantage in the average over equally weighted classes. For example, count features yield nonzero F1 on environment, while TF-IDF misses all five test documents in that class. Figure 5 summarizes the accuracy–Macro-F1 gap for all seven methods.

The leading methods make complementary errors. Comparing their saved predictions, Word Frequency and TF-IDF are both correct on 1,209 documents and both wrong on 68. Word Frequency alone is correct on 10, whereas TF-IDF alone is correct on 15. Thus, TF-IDF's five additional correct predictions are a net difference, not uniform superiority on every document. On the 12 low-frequency classes, their Macro-F1 scores are 76.48% and 73.71%, respectively. This supports choosing between them according to the evaluation objective, rather than treating the 0.38-point accuracy advantage as a definitive ranking.

Why weighting can help without helping every class. Repeated topic terms can strengthen count-based evidence, whereas TF-IDF reduces ubiquitous-word contributions and normalizes the document scale. These changes can benefit broad topical discrimination while changing the precision–recall balance for small classes. However, the TF-IDF comparison changes both IDF weighting and L2 normalization, and each representation selects its own C. The experiment identifies a useful complete pipeline; it does not isolate which of these components causes each improvement.

Word Frequency assigns five international-business documents and five economy documents to business. It assigns four environment documents and three cosmos documents to science. Its confusion matrix in Figure 7 shows strong diagonal performance for the large sports categories, with remaining errors concentrated in cross-topic decisions. These errors explain how a lexical model can achieve high accuracy while retaining imperfect separation among business, policy and science topics.

The strong sparse baseline suggests that explicit topic vocabulary supplies substantial useful evidence in this corpus: retaining word order is not necessary to reach a high score here. It does not establish that word order is unimportant for language understanding. Closely related labels may share vocabulary, and tasks involving negation, relations or sentiment can require information discarded by bag of words. For this coursework, the lesson is to establish a competitive lexical baseline before attributing gains or failures to model complexity.

4 Task 2: 100-Dimensional Word Embeddings

Task 2 completes the three required embedding experiments: pretrained GloVe 6B, Word2Vec trained on AG News texts, and Word2Vec trained on NYT training texts. Every source uses 100-dimensional word vectors, ordinary mean pooling over known tokens, and logistic regression.

4.1 Embedding Sources and Training

Pretrained GloVe 6B. GloVe learns word vectors from global co-occurrence statistics (Pennington et al., 2014). The experiment uses the released `glove.6B.100d` vectors, trained on Wikipedia 2014 and Gigaword 5. The original release contains 400,000 words. Loading retains the 70,193 entries that occur in the tokenized NYT corpus. These vectors remain fixed during NYT classification.

Word2Vec trained on AG News. A 100-dimensional skip-gram model learns local word-context relationships (Mikolov et al., 2013) from the 120,000 AG News training texts. Labels are unused, and the resulting vectors represent NYT documents for classification.

Word2Vec trained on NYT. A second 100-dimensional skip-gram model uses the 10,405 NYT training documents. Validation and test documents contribute no word-context pairs during embedding training. Its vectors are used in the same document-classification pipeline.

Both Word2Vec models use a context window of five, a minimum token count of two, five negative samples, frequent-word subsampling at 10^{-3}, ten epochs, one worker, and seed 42. Their vocabularies contain 43,594 and 54,598 words for AG News and NYT, respectively.

Skip-gram with negative sampling encourages observed word–context pairs to receive larger dot products than sampled alternatives. For a target word w, observed context c, and five negative

contexts n_j, a training contribution has the form

$$\log \sigma(u_c^\top v_w) + \sum_{j=1}^{5} \log \sigma(-u_{n_j}^\top v_w), \tag{5}$$

where v and u are target and context embeddings. The classifier uses the learned target-word vectors from Gensim. Shared settings make the source corpus the principal experimental difference between the two Word2Vec representations.

4.2 Mean Pooling and Document Classification

For each embedding source, a document is represented by the mean of its known token vectors. Let K_d be the multiset of token occurrences with available embeddings. The document vector is

$$e(d) = \frac{1}{|K_d|} \sum_{w \in K_d} e(w), \qquad e(d) \in \mathbb{R}^{100}. \tag{6}$$

Repeated tokens contribute repeatedly. Unknown tokens are excluded from both the sum and its denominator. If K_d is empty, the vector is zero. This averaging operation discards token order and compresses all known words into 100 values, allowing semantically similar words to share evidence through nearby vectors.

Each of the 100 feature coordinates is standardized using its training mean and standard deviation, with the same transformation applied to validation and test vectors. Word vectors remain fixed during logistic-regression training. The C search and validation selection follow Task 1. Figure 1 shows the full embedding-to-classification pipeline.

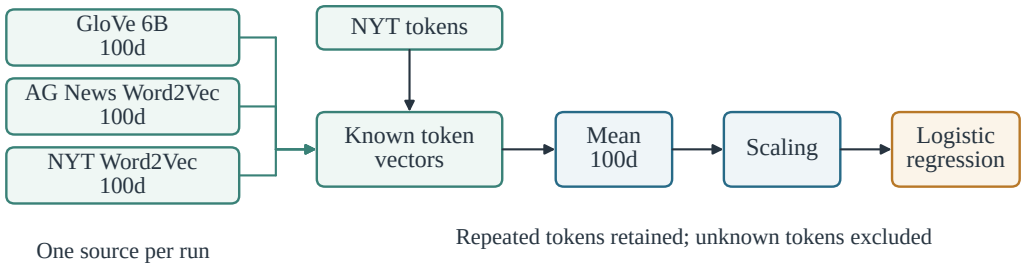


Figure 1: Task 2 embedding sources and shared classification pipeline. Each run uses one source. Every known token occurrence contributes to the mean; unknown tokens are excluded from the sum and denominator. Scaling and logistic regression are fitted on training data.

4.3 Results and Vocabulary Coverage

Table 2 reports the three required embedding experiments. The best averaged embedding is NYT Word2Vec, with 81.85% Macro-F1. It approaches the sparse models despite reducing every document to 100 features, compared with 80,350 lexical coordinates. GloVe and AG News Word2Vec produce lower scores. A dense representation can share information across words, but its average cannot retain separate indicators for every discriminative term or reconstruct token order.

Table 2: Task 2 validation and test results (%). All sources use 100-dimensional vectors, ordinary mean pooling and logistic regression.

	Validation		Test		
Method	Acc.	Macro-F1	Acc.	Macro-F1	Selection
GloVe	92.92	83.72	91.71	79.38	$C = 0.1$
Word2Vec (AG News)	91.38	76.81	90.02	77.11	$C = 0.1$
Word2Vec (NYT)	93.77	83.33	92.63	81.85	$C = 1$

In the main run, NYT Word2Vec exceeds AG News Word2Vec by 4.74 Macro-F1 points. The two models share dimensions, training settings, mean pooling, and classifier search. Table 3 reports occurrence-weighted coverage on the test set: 99.08% for NYT Word2Vec, 97.10% for AG News Word2Vec, and 98.05% for GloVe.

Table 3: Embedding vocabularies and occurrence-weighted coverage on NYT test documents. GloVe's vocabulary size refers to the full pretrained release. Every test document has at least one known token for all three sources.

Embedding	Vocabulary	Test coverage (%)	Empty documents
GloVe	400,000	98.05	0
Word2Vec (AG News)	43,594	97.10	0
Word2Vec (NYT)	54,598	99.08	0

The NYT embedding vocabulary is smaller than GloVe's original vocabulary, yet it covers a larger fraction of test-token occurrences. Vocabulary size alone therefore does not describe how well a resource matches this task. The advantage is consistent with the local NYT corpus supplying more relevant vocabulary and word-context statistics. Every test document contains known tokens under all three embedding vocabularies.

Domain relevance is not the same as corpus size. AG News supplies 120,000 training texts, yet the vectors learned from 10,405 NYT training documents perform better on NYT. This is consistent with task-relevant co-occurrence statistics and vocabulary being more useful than the number of source documents alone. The comparison is not a controlled scaling experiment: source corpora also differ in document length and topic composition, and coverage does not measure vector quality. Likewise, a covered word can still have an unhelpful representation for a fine-grained topic. We therefore interpret the result as evidence for in-domain representation learning, not proof that smaller corpora or Word2Vec are generally superior to larger pretrained resources.

Compression trades specificity for shared semantics. NYT mean embeddings are only 1.76 Macro-F1 points below Word Frequency while using 100 document coordinates instead of 80,350. This is a useful compact representation, but two documents with different discriminative words can have nearby averages. Once these distinctions have been pooled away, a linear classifier cannot recover them. The coordinate reduction is not a measurement of whole-model memory or deployment cost, because embedding tables and preparation remain necessary. The pooling controls below test whether better weighting can reduce, rather than eliminate, this information loss.

5 Task 3: Pretrained BERT Fine-Tuning

Task 3 fine-tunes `google-bert/bert-base-uncased` for end-to-end text classification. The required setting is `max_length=64` and exactly **3 epochs**, followed by Accuracy and Macro-F1 evaluation on the shared NYT test set. This section reports that prescribed configuration; other input lengths, class weighting and training budgets appear in Section 6.

5.1 Architecture and Tokenization

BERT supplies contextual token representations from a pretrained bidirectional Transformer (Devlin et al., 2019). The experiment uses `google-bert/bert-base-uncased`, with 12 encoder layers, hidden size 768, and 12 attention heads. Its uncased WordPiece tokenizer processes the original text. The main setting truncates and pads inputs to a total length of 64, including the special tokens. Attention masks distinguish content from padding.

Figure 2 depicts the implemented `BertForSequenceClassification`. The final-layer `[CLS]` state passes through a 768-dimensional Dense layer and tanh pooler, then dropout with probability 0.1 and a $768 \rightarrow 29$ linear head. The largest logit determines the predicted class. Cross-entropy training updates the encoder, pooler and head jointly. Each encoder layer contains multi-head self-attention, residual connections, LayerNorm and a GELU feed-forward network.

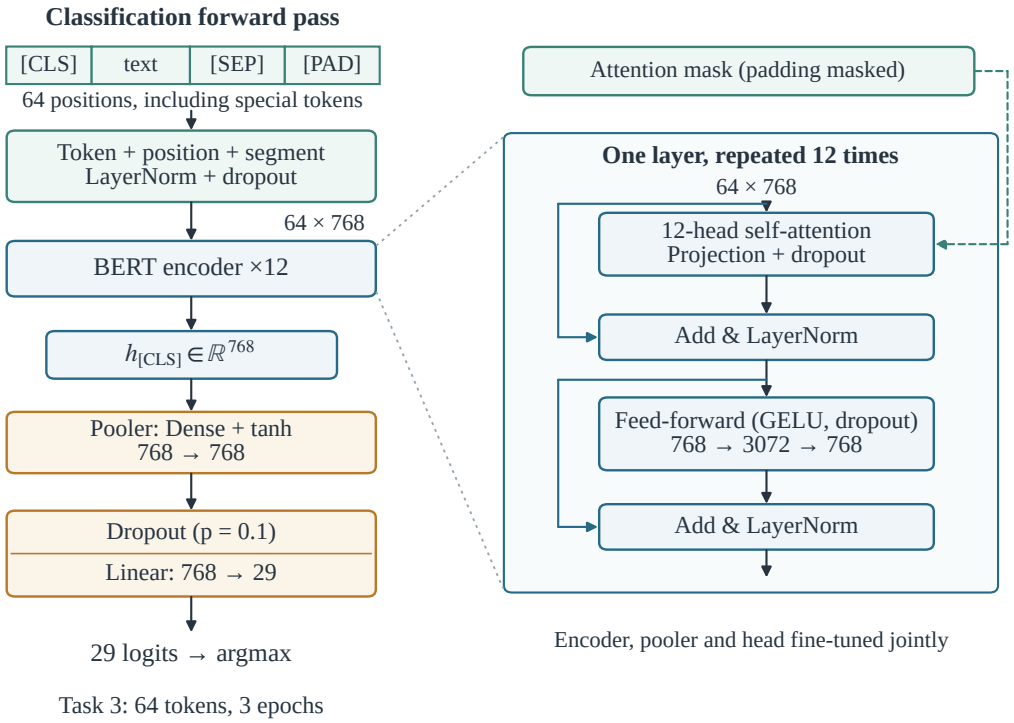


Figure 2: BERT classification architecture for Task 3. Left: the complete forward pass. Right: one encoder layer repeated 12 times. The attention mask excludes padding; gray dotted lines indicate the expanded layer detail. The 64-position budget includes special tokens and padding, leaving at most 62 text WordPieces.

5.2 Fine-Tuning and Checkpoint Selection

BERT uses AdamW with a learning rate of 2×10^{-5}, training batch size 32, evaluation batch size 128, and weight decay 0.01. Biases and layer-normalization weights are excluded from weight decay. The learning rate increases linearly during the first 10% of training steps and then decreases linearly to zero. Gradients are clipped to norm 1.0. Training uses bfloat16 automatic mixed precision on an NVIDIA H100 80GB GPU.

Training runs for exactly three epochs, with validation evaluation after each epoch. The checkpoint with the highest validation Macro-F1 is saved, reloaded and evaluated on the test set. Epoch three is selected in this run.

5.3 Required-Setting Results and Learning Curves

Table 4 reports the prescribed BERT setting. BERT reaches 89.63% test accuracy and 62.97% Macro-F1. Figure 3 shows that training loss decreases across all three epochs, while validation Macro-F1 rises from 48.82% to 61.03% and then 63.37%. Validation accuracy rises from 85.77% to 89.08%. The selected checkpoint is therefore the final epoch, and the measured validation trajectory is still improving at the prescribed stopping point.

Table 4: Task 3 validation and test results (%): BERT-base-uncased, length 64, three epochs and uniform cross-entropy.

	Validation		Test		
Method	Acc.	Macro-F1	Acc.	Macro-F1	Selection
BERT	89.08	63.37	89.63	62.97	epoch 3

The input budget provides a plausible explanation for part of the difference from full-document methods. NYT documents have a median length of 686 regex-tokenized words, whereas a BERT input of 64 WordPiece tokens leaves at most 62 positions for text after special tokens. The tokenization units differ, but the prefix constraint is much shorter than a typical document. Sparse features and mean embeddings can aggregate topic evidence throughout the text; BERT can use contextual relationships only within the retained prefix.

The seven zero-F1 categories also motivate the class-weighting study. BERT learns a new classification head during three epochs, and the unweighted main setting gives frequent classes more training examples. The following experiments vary input access and the loss objective directly.

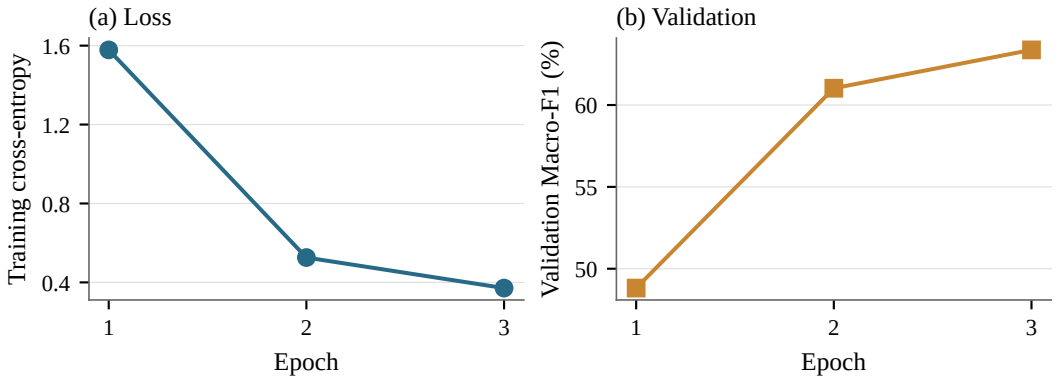


Figure 3: BERT training loss and validation Macro-F1 in the main three-epoch run. Loss is the example-weighted mean cross-entropy; validation selection retains epoch three.

5.4 Class-Level Performance and Error Analysis

Table 7 contrasts a frequent sports class, a business class, and several small classes. Baseball is classified well by all three displayed models. BERT exceeds both sparse models on international business, showing that its lower overall Macro-F1 is not uniform across categories. For abortion and surveillance, the sparse models obtain 100% F1, while BERT obtains zero. Those scores involve only three test documents per class.

BERT has zero test F1 in 7 classes: abortion, cosmos, environment, federal budget, gun control, immigration, and surveillance. These classes have between 22 and 89 training documents. Frequent categories contribute many more documents to accuracy, while each of these seven classes contributes one twenty-ninth of Macro-F1. Their zero scores directly lower the macro average. The full per-class heatmap in Figure 6 visualizes this pattern across all seven methods.

Saved predictions reveal a more specific failure: BERT never predicts any of these seven labels on the reference test set. Together they contain only 38 documents, or 2.92% of the test set, but account for 24.14% of Macro-F1's class terms. This is label suppression, not merely a small decrease shared by every category. Across all 12 low-frequency classes, main-run BERT reaches 30.55% Macro-F1, versus 76.48% for Word Frequency. These diagnostics make minority-class learning a separate hypothesis to test alongside insufficient text access.

BERT assigns all five test documents from each of cosmos and environment to science, confusing specialized topics with a broader related topic. The original label inventory retains both broad and specialized categories, and evaluation uses exact multiclass matching.

Concrete errors challenge a truncation-only explanation. Cleaned document `row_id=8195` is labeled abortion and contains "abortions" within BERT's retained 62 text WordPieces. Both count and TF-IDF classify it correctly, but BERT predicts the Affordable Care Act. Document `row_id=8527` is labeled cosmos; "NASA" and "Kepler" already occur in the retained prefix, yet BERT predicts science while both sparse models are correct. Important lexical cues are therefore not always missing from its input. These examples are compatible with insufficient minority-class

learning or weak fine-grained decision boundaries, which the loss and budget controls investigate; keyword presence alone is not a causal attribution.

The errors are also complementary. On document `row_id=2143`, BERT correctly predicts international business while both sparse models predict business. Its international-business recall is 84.38%, compared with 71.88% for Word Frequency. A lower aggregate score can coexist with useful class-specific behavior. These selected examples illustrate error patterns that motivate further controls; they are not a representative estimate of all errors.

6 Supplementary Exploratory Experiments

Beyond the three required tasks, supplementary experiments examine input length, loss weighting, pooling, repeated partitions, training budget and text position, bringing the study to 233 distinct configurations. The fixed partition uses three training seeds; the repeated study uses five partition seeds. Validation Macro-F1 selects every model. Appendix C gives the complete design, and Appendices D and E give configuration-level results.

Class weighting uses training frequencies, $w_k = N_{\text{train}}/(29N_k)$. Low-frequency F1 averages classes with at most 100 training documents in each partition. Across-partition summaries first average training seeds within each partition, then report the mean and sample standard deviation of the five partition means. Fixed-partition error bars describe the three training seeds.

6.1 Context Length and Class Weighting

Longer inputs improve BERT under the same three-epoch budget (Table 5; Figure 4). Uniform-loss Macro-F1 rises from $65.60 \pm 3.27\%$ at 64 tokens to $76.62 \pm 1.74\%$ at 256, a mean gain of 11.03 points. All three paired seeds improve, by 7.92 to 13.45 points. This gain accompanies substantial truncation: the corpus median is 890 WordPiece tokens, and 99.95% of documents exceed the 62-token text budget. That budget retains a median of 49 regex-tokenized words.

Table 5: BERT context and loss study. Scores are percentages, shown as mean ± sample standard deviation over three training seeds. Low-frequency F1 averages classes with at most 100 training documents. Every run trains for three epochs.

Length	Loss	Accuracy	Macro-F1	Rare F1
64	Uniform	89.96 ± 0.45	65.60 ± 3.27	36.24 ± 7.11
64	Weighted	89.68 ± 0.19	79.53 ± 1.46	72.85 ± 2.85
128	Uniform	92.11 ± 0.38	72.38 ± 2.89	50.50 ± 7.72
128	Weighted	91.81 ± 0.12	82.39 ± 0.27	75.86 ± 0.76
256	Uniform	93.75 ± 0.18	76.62 ± 1.74	58.31 ± 4.26
256	Weighted	92.93 ± 0.08	83.26 ± 0.47	76.67 ± 1.25
512	Uniform	94.19 ± 0.23	80.15 ± 1.74	66.43 ± 4.73
512	Weighted	93.45 ± 0.12	84.45 ± 0.59	78.43 ± 1.42

Class weighting primarily improves low-frequency topic recognition. At 64 tokens, mean low-frequency F1 rises from 36.24% to 72.85%; at 256, it rises from 58.31% to 76.67%. At lengths 64–256, every weighted run has nonzero F1 in all 29 classes. At 256, weighting reduces accuracy by 0.82 points and raises Macro-F1 by 6.63 points. Weighted BERT reaches $83.26 \pm 0.47\%$ at 256 and $84.45 \pm 0.59\%$ at 512. Even the 512-token budget truncates 81.16% of documents, retaining an average 64.45% of their WordPieces. Longer context therefore increases available evidence while preserving a substantial prefix constraint.

Reweighting changes the learning objective. Before normalizing weighted loss within each batch, $N_k w_k = N_{\text{train}}/29$ equalizes the aggregate training weight of each class. The intended effect is to make small-class errors matter more during optimization, not to supply additional text. At a fixed length of 64, weighting raises mean Macro-F1 by 13.93 points and low-frequency F1 from 36.24% to 72.85%, while accuracy decreases by only 0.28 points. This directly supports treating class imbalance as a limitation distinct from truncation. Weighted cross-entropy is still a surro-

gate: it does not directly optimize the non-decomposable Macro-F1 metric or guarantee improved precision for every class.

The context and weighting gains should not be added as independent effects. Weighting's Macro-F1 advantage is 13.93 points at 64 tokens but 6.63 at 256; extending context improves ordinary loss by 11.03 points and weighted loss by 3.73. Both interventions help small topics, so their benefits overlap under this three-epoch protocol. Longer inputs also increase attention computation and memory demand. The more useful conclusion is to jointly examine text access and the training objective, rather than automatically choosing the largest length.

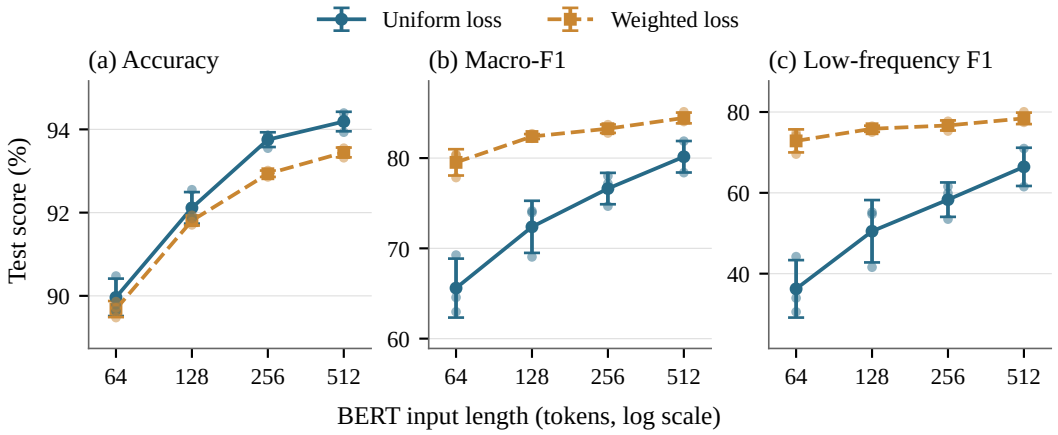


Figure 4: BERT input-length and loss comparison: (a) accuracy, (b) Macro-F1, and (c) low-frequency F1. Faint points are individual runs; lines and error bars are three-seed means and sample standard deviations. Weighting improves low-frequency recognition at each input length.

6.2 Matched Text Access for Sparse Models

Table 12 isolates text access within each sparse representation. At 64 tokens, TF-IDF reaches 73.47% Macro-F1 and Word Frequency reaches 71.31%, compared with 82.85% and 83.61% on full documents. Count and TF-IDF improve with each larger budget; binary features decline at 128 before improving at 256. The training prefixes cover the same characters as BERT's retained text, exposing the value of later topic evidence within each classifier pipeline.

At the 64-token span, TF-IDF's 73.47% Macro-F1 still exceeds ordinary BERT's three-seed mean of 65.60%. Limited text access therefore explains part, but not all, of the main BERT deficit. Weighted BERT at that length reaches 79.53%, indicating that useful discrimination can be learned even from the short prefix. The latter comparison also changes the loss objective; it is not an architecture-only experiment. Equal visible spans are an important control, but fair representation comparisons must additionally state the objective, validation selection and optimization budget.

6.3 Pooling and Word2Vec Training Variation

IDF weighting improves the document representation for each embedding source (Table 13, Appendix D). Mean Macro-F1 rises by 0.54, 1.36, and 1.00 points for GloVe, AG News Word2Vec, and NYT Word2Vec. The pooling operation retains repeated occurrences and reduces the relative contribution of common words. Across three seeds, NYT Word2Vec exceeds AG News Word2Vec by 3.56 points under mean pooling. Each pooling pair uses the same embeddings, dimension, and classifier search. Figure 8 summarizes both the matched-prefix and pooling comparisons.

IDF pooling is motivated by a representation bottleneck: frequent generic words occupy much of an ordinary mean, whereas less ubiquitous terms can carry topic-specific evidence. Reweighting changes their relative contributions without retraining vectors or increasing the 100-dimensional document size. The effect is modest, and not uniform across seeds: NYT's seed-42 Macro-F1 changes by -0.01 points, while seeds 43 and 44 gain 1.71 and 1.30 points. AG News improves for

all three seeds. Thus the higher mean is useful evidence for the pooling choice, but three runs do not establish universal improvement or statistical significance. The average still discards word order and can dilute a rare decisive term.

6.4 Ranking Across Partitions and Loss Objectives

Table 18 compares full-document classical models with the longer-context BERT configurations across five partitions. Weighted BERT-512 reaches $84.22 \pm 0.87\%$ Macro-F1, compared with 81.68% for ordinary Word Frequency and 83.35% for its weighted counterpart. Its paired mean differences are +2.53 and +0.87 points, respectively. Figure 10 displays every partition mean and the paired differences.

Weighted TF-IDF reaches $84.49 \pm 2.75\%$ Macro-F1; weighted BERT-512 has a paired mean difference of -0.27 points relative to it and less variation across partitions. The ranking therefore changes when lexical classifiers receive the same class-weighting rule.

Figure 9 applies the same inverse-frequency rule to each classifier family. At 512 tokens, weighting changes BERT's Macro-F1 by 4.10 points and accuracy by -0.92 points. NYT Word2Vec-IDF instead falls from 81.07% to 80.43% Macro-F1 under weighting. Figure 13 resolves the class-level changes. Military loses F1 under weighting at both lengths on the reference partition. Extending weighted BERT from 256 to 512 adds 0.77 points across partition means. Table 25 gives the repeated-partition sparse-prefix controls.

The paired results sharpen what the ranking supports. Weighted BERT-512 exceeds ordinary Word Frequency in all five partitions, with differences from $+0.84$ to $+4.03$ points. Against weighted TF-IDF, however, it wins only two partitions and loses three; the paired difference is -0.27 ± 2.03 points (mean $\pm$ sample standard deviation). Neither of these two methods wins consistently across the observed partitions, and the standard deviation is not a confidence interval. A claim that BERT generally dominates the lexical baseline would therefore depend on which loss was allowed for the baseline.

Weighting is also not a universal cure. Its negative mean effect on NYT Word2Vec-IDF suggests that shifting decision boundaries may fail to resolve overlapping pooled representations and can introduce additional false positives. This is a hypothesis, not a diagnosed cause: class-level confusion changes and within-/between-class feature distances would be needed to test it. The observed counterexample is enough to reject the blanket rule that inverse-frequency weighting must improve Macro-F1.

6.5 Optimization Budget and Evidence Location

With the same 64-token prefix, ten-epoch uniform-loss BERT reaches 79.80% Macro-F1, changing the score by +14.20 points. At 512, ordinary loss reaches 86.38%, compared with 85.72% for weighting. Ordinary loss also leads at 256 under this budget, reversing the three-epoch loss ordering. Table 26 and Figure 11 give the selected checkpoints and validation trajectories. At a fixed weighted BERT-256 budget, head-only inputs achieve 83.26%, head + tail 81.57%, and tail-only 76.22% (Table 27; Figure 12). Retaining the full prefix is strongest among these spans on the reference partition.

Training duration changes the interpretation of weighting. Keeping length 64 fixed, ten-epoch ordinary loss raises Macro-F1 by 14.20 points and low-frequency F1 from 36.24% to 72.81%. No later text has been added, so input truncation cannot be the only bottleneck. Weighting may help small classes learn earlier, whereas more optimization lets the ordinary objective recover much of that deficit. This interpretation is consistent with the observed scores, not a direct measurement of per-class gradient dynamics. The ten-epoch run also changes the warmup/decay horizon, so the effect cannot be attributed solely to the number of additional epochs.

The ten-epoch loss reversal concerns means, not demonstrated statistical superiority: ordinary loss leads by only 1.02 points at 256 and 0.66 at 512 over three seeds. Weighted BERT-256 selects epoch four for all three seeds, although all ten epochs are completed; ordinary-loss selections are epochs six, nine and four. A longer allowed budget and validation selection should therefore be distinguished from using the last checkpoint. The trajectories motivate monitoring minority-sensitive

validation performance rather than concluding that more epochs or permanent reweighting is always better.

Evidence location matters as well as its amount. The head exceeds the tail by 7.03 Macro-F1 points, and replacing half the head with the tail lowers the mean by 1.68 points. This is compatible with news leads carrying concentrated topic cues. It does not show that endings are useless: tail-only still reaches 76.22%, while the head–tail control simultaneously removes part of the lead and changes textual continuity. A future sliding-window or hierarchical aggregation comparison could test access to later evidence without making that trade-off.

7 Discussion: Lessons from the Exploration

The main result is not a hierarchy in which a more complex representation must be better. Full-document lexical features are strong for this news task; in-domain averaging is competitive with a compact document representation; and BERT's apparent weakness changes when the input, objective and budget are controlled. The exploratory value lies in turning an unexpected score into separate, testable hypotheses. Prefix controls probe access to evidence, pooling controls probe how that evidence is compressed, and loss/budget controls probe how classes are learned. No single intervention explains every observed difference.

8 Limitations

The five partitions reuse one corpus and have overlapping documents. Their standard deviations describe partition sensitivity, not confidence intervals for an independent population. Some classes have three to five test documents in the reference partition, making their F1 sensitive to a few predictions. Partition 43 has no cosmos test documents; its F1 is zero in the fixed 29-class macro average. Low-frequency class membership is determined separately from each training partition. The extensions were designed after inspecting the main comparison, so their test results are exploratory. The label inventory mixes broad and specialized topics under flat multiclass evaluation.

BERT accesses at most 512 tokens, and the ten-epoch and position studies use one partition. Concatenating head and tail also changes local continuity. Matched spans undergo representation-specific tokenization and fitting. Word2Vec source comparisons jointly change corpus content, coverage, and document-length distributions. Elapsed times depend on preparation, hardware, and concurrent tasks.

9 Conclusion

Task 1 yields the highest accuracy with TF-IDF (94.01%) and the highest Macro-F1 with Word Frequency (83.61%). In Task 2, NYT Word2Vec leads the averaged embeddings at 81.85% Macro-F1. Task 3 completes the prescribed 64-token, three-epoch BERT fine-tuning, obtaining 89.63% accuracy and 62.97% Macro-F1. The supplementary experiments show that text access, class weighting and training budget affect representation rankings. On the reference partition, ordinary loss has higher mean Macro-F1 than weighted loss at lengths 256 and 512 under the ten-epoch budget; this is not a claim of significant superiority across corpora.

The exploration's central lesson is to explain a ranking before generalizing it. BERT's minority-label suppression cannot be reduced to missing later text alone, and its recovery under different objectives and budgets does not erase the value of the lexical baseline. Useful conclusions pair scores with class-level errors, controlled comparisons and explicit uncertainty. Proposed follow-up experiments should validate these explanations rather than merely seek a higher test score.

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A Class Distribution and Detailed Results

A.1 Complete Seven-Method Comparison

Table 6: Validation and test performance (%) of all seven methods across the three tasks on the common partition. Selection uses validation Macro-F1; bold marks the best test score. All methods except BERT use logistic regression.

	Validation		Test		
Method	Acc.	Macro-F1	Acc.	Macro-F1	Selection
Binary BoW	94.38	84.86	92.47	80.47	$C = 10$
Word Frequency	93.69	84.02	93.63	**83.61**	$C = 1$
TF-IDF	95.00	84.86	**94.01**	82.85	$C = 10$
GloVe	92.92	83.72	91.71	79.38	$C = 0.1$
Word2Vec (AG News)	91.38	76.81	90.02	77.11	$C = 0.1$
Word2Vec (NYT)	93.77	83.33	92.63	81.85	$C = 1$
BERT	89.08	63.37	89.63	62.97	epoch 3

A.2 Class Distribution and Class-Level Results

Table 8 lists the counts in the shared partition. Class IDs follow the alphabetical label order used by the implementation. Spaces in the displayed names replace underscores in the original labels. The same IDs index Figure 7.

Table 7: Selected class-level test F1 scores and split sizes in the main comparison. Count denotes Word Frequency.

	Documents		Test F1 (%)		
Class	Train	Test	Count	TF-IDF	BERT
baseball	1795	212	99.05	99.05	97.39
international business	384	32	71.88	72.13	80.60
abortion	22	3	100.00	100.00	0.00
cosmos	30	5	44.44	40.00	0.00
environment	40	5	28.57	0.00	0.00
surveillance	40	3	100.00	100.00	0.00

Table 8: NYT class counts after conflicting-label removal and exact-text deduplication.

ID	Class	Train	Validation	Test
1	abortion	22	3	3
2	baseball	1795	208	212
3	basketball	792	95	110
4	business	351	38	51
5	cosmos	30	6	5
6	dance	87	8	13
7	economy	209	29	32
8	energy companies	115	14	9
9	environment	40	4	5
10	federal budget	89	14	6
11	football	1226	146	160
12	gay rights	68	9	7
13	golf	799	114	114
14	gun control	46	5	7
15	hockey	513	66	75
16	immigration	44	5	9
17	international business	384	42	32
18	law enforcement	331	41	41
19	military	53	5	3
20	movies	265	29	34
21	music	326	51	36
22	politics	372	44	42
23	science	336	49	43
24	soccer	1081	133	141
25	stocks and bonds	91	9	10
26	surveillance	40	6	3
27	television	159	24	11
28	tennis	652	88	77
29	the affordable care act	89	15	11
	Total	10,405	1,300	1,302

Table 9 reports all per-class F1 scores. AG and NYT denote the two Word2Vec sources. Binary, Count, and TF-IDF use the shared NYT training vocabulary. Figure 6 displays the same measurements as a heatmap.

Table 9: Complete test F1 scores (%) by class and method.

ID	Class	Binary	Count	TF-IDF	GloVe	AG	NYT	BERT
1	abortion	80.00	100.00	100.00	66.67	80.00	100.00	0.00
2	baseball	98.34	99.05	99.05	98.57	97.85	98.57	97.39
3	basketball	98.62	99.08	99.54	95.81	94.84	97.70	91.67
4	business	67.33	75.00	77.36	75.73	68.04	77.23	76.47
5	cosmos	57.14	44.44	40.00	36.36	40.00	44.44	0.00
6	dance	96.00	96.00	96.00	92.31	86.96	96.00	96.00
7	economy	69.09	77.19	77.97	79.31	73.68	79.31	74.29
8	energy companies	60.00	57.14	60.87	50.00	40.00	42.11	57.14
9	environment	0.00	28.57	0.00	33.33	57.14	57.14	0.00
10	federal budget	57.14	57.14	61.54	50.00	57.14	37.50	0.00
11	football	98.76	98.77	99.38	96.64	96.32	97.55	92.83
12	gay rights	76.92	72.73	83.33	76.92	83.33	72.73	66.67
13	golf	98.67	99.12	100.00	98.21	98.23	100.00	97.37
14	gun control	72.73	92.31	92.31	92.31	92.31	92.31	0.00
15	hockey	99.34	100.00	100.00	100.00	98.68	100.00	99.33
16	immigration	80.00	82.35	82.35	82.35	66.67	70.59	0.00
17	international business	71.88	71.88	72.13	72.73	69.44	70.59	80.60
18	law enforcement	83.72	87.50	86.42	81.01	79.01	82.35	69.57
19	military	80.00	85.71	80.00	85.71	75.00	85.71	60.00
20	movies	95.77	97.14	94.44	94.44	94.29	89.55	89.86
21	music	93.33	95.77	100.00	95.77	87.67	93.15	94.12
22	politics	80.90	82.35	83.15	72.73	69.66	75.29	78.26
23	science	84.54	83.87	83.52	83.15	83.52	87.91	81.63
24	soccer	97.90	98.59	98.59	97.18	97.14	97.87	98.59
25	stocks and bonds	76.19	88.89	82.35	84.21	72.73	88.89	82.35
26	surveillance	100.00	100.00	100.00	85.71	75.00	85.71	0.00
27	television	90.00	85.71	85.71	76.19	64.00	83.33	81.82
28	tennis	99.35	98.72	100.00	98.72	97.44	100.00	98.72
29	the affordable care act	70.00	69.57	66.67	50.00	40.00	70.00	61.54

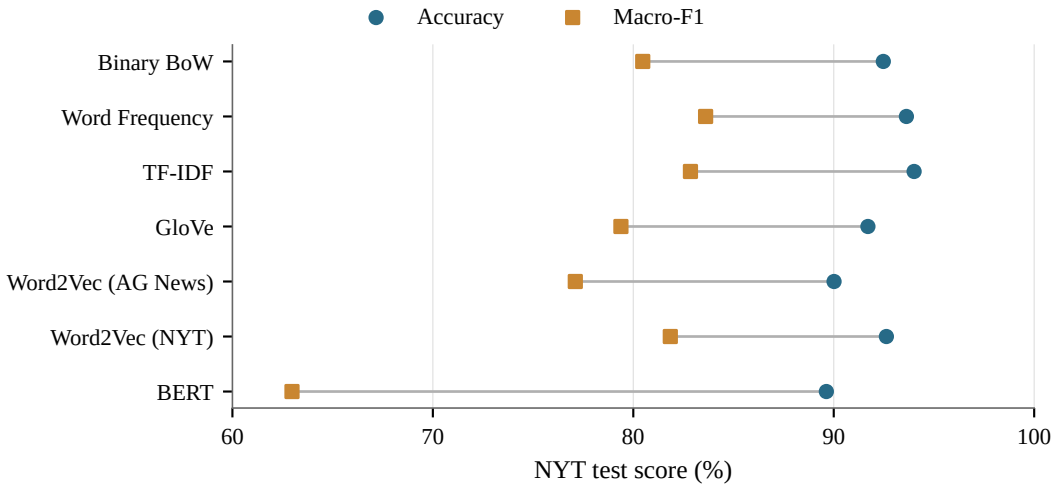


Figure 5: Main-run accuracy and Macro-F1 for the seven methods. Connecting segments show the gap between document-weighted accuracy and equally weighted class F1. Exact scores and model selections appear in Table 6.

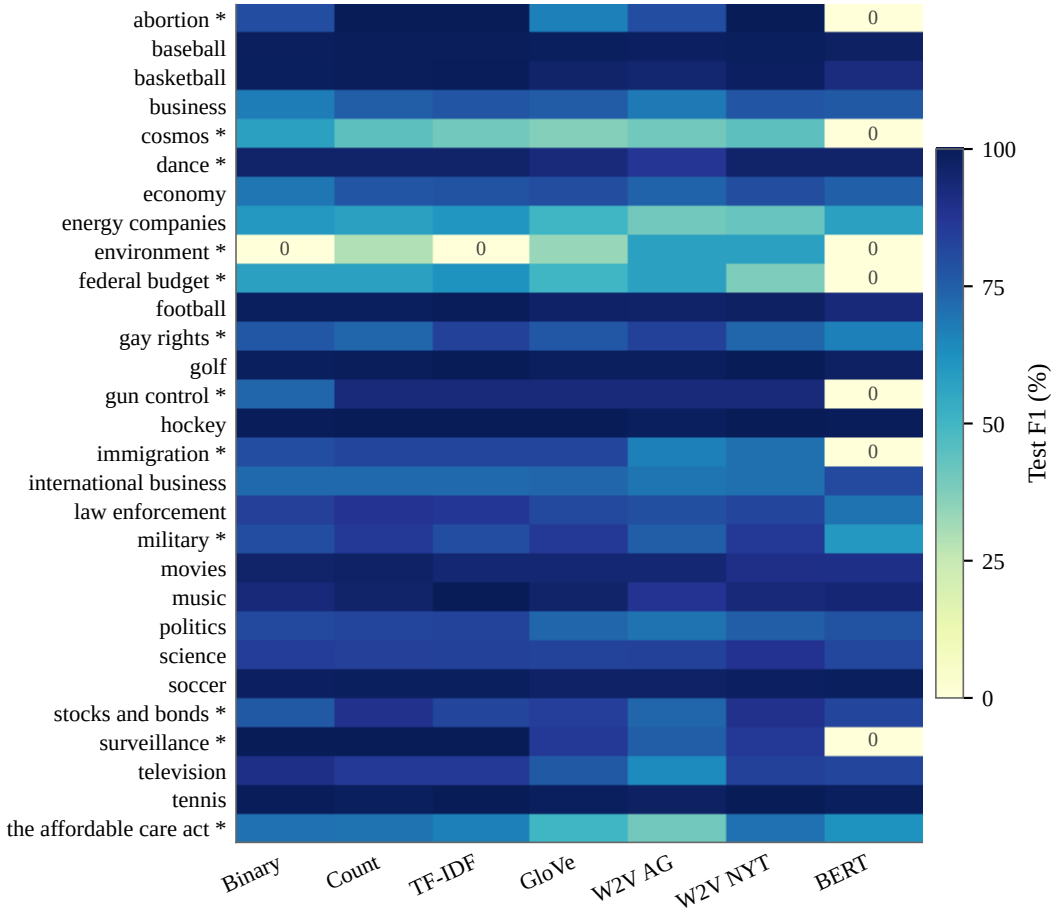


Figure 6: Per-class test F1 in the main comparison. Asterisks mark classes with at most 100 training documents; zero-F1 cells are labeled. Count denotes Word Frequency, and W2V AG/NYT denotes the Word2Vec source. Darker cells have higher F1.

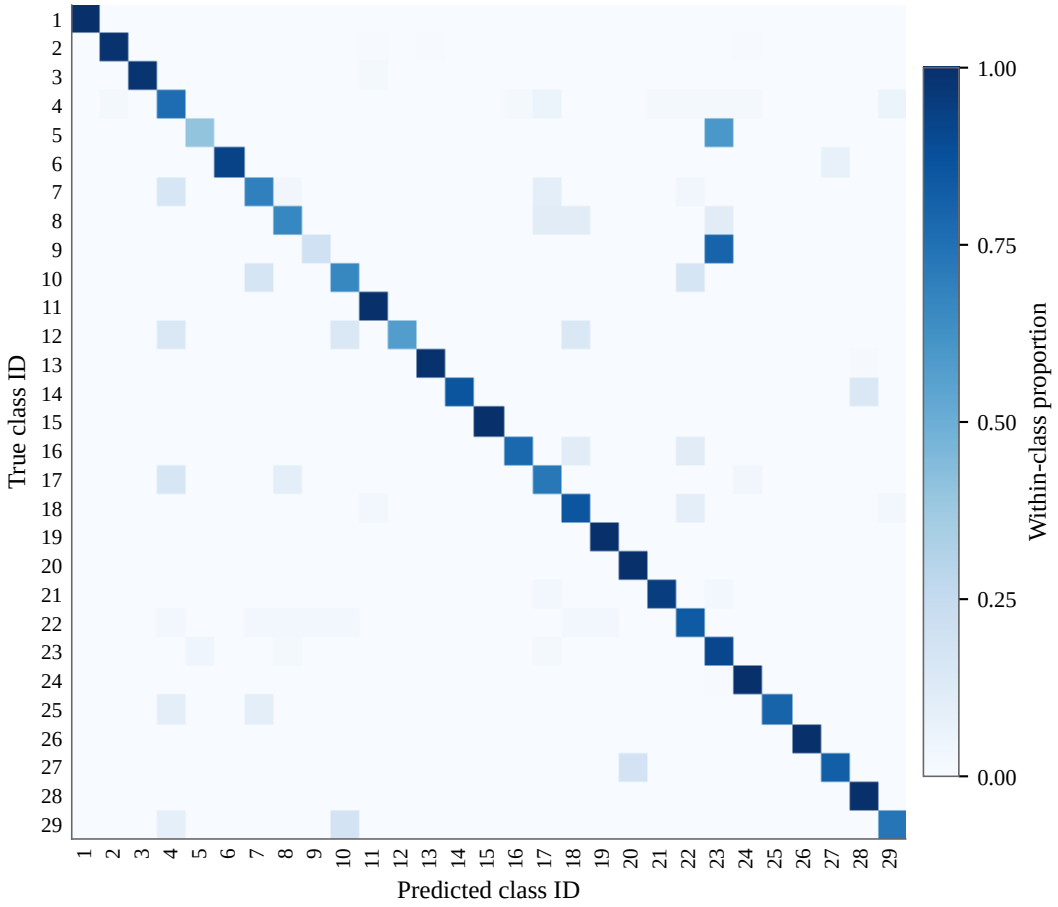


Figure 7: Word Frequency test confusion matrix, normalized within each true class. Rows are true labels and columns are predicted labels. Class IDs are defined in Table 8. Each row sums to one.

B TRAINING RECORDS AND REPRODUCIBILITY

Table 10 gives the saved BERT epoch measurements. Table 11 gives elapsed times measured within each method. Traditional-method timing includes representation preparation, three logistic regression trials, evaluation, and output writing. Word2Vec timing also includes embedding training. BERT timing includes model loading, fine-tuning, checkpoint writing and loading, and evaluation. The 768-dimensional BERT entry refers to its contextual hidden representation, not a fixed feature matrix used by logistic regression.

Table 10: BERT training history and validation metrics.

Epoch	Training loss	Validation Acc. (%)	Validation Macro-F1 (%)
1	1.5785	85.77	48.82
2	0.5259	88.46	61.03
3	0.3713	89.08	63.37

Table 11: Observed elapsed times and representation dimensions. CPU and GPU times use different hardware and include different preparation operations.

Method	Feature dimension	Elapsed time (s)	Device
Binary BoW	80350	60.16	CPU
Word Frequency	80350	383.80	CPU
TF-IDF	80350	64.84	CPU
GloVe	100	20.29	CPU
Word2Vec (AG News)	100	192.22	CPU
Word2Vec (NYT)	100	277.55	CPU
BERT	768	70.11	H100 GPU

B.1 Execution and Saved Evidence

The `homework1/README.md` specifies environment installation. From the repository root, enter `homework1/` and run the following commands to download resources, execute experiments, and recompute metrics from predictions. The repository publishes the final PDFs, not report-generation tools:

```
cd homework1
python scripts/prepare_assets.py
PYTHONHASHSEED=42 python run_experiments.py \
  --bert-model data/bert-base-uncased
PYTHONHASHSEED=42 python -m scripts.run_explorations
PYTHONHASHSEED=42 python -m scripts.run_scaling
python scripts/verify_results.py --results outputs/main \
  --explorations outputs/explorations
python scripts/verify_scaling.py
```

Running the experiments creates the output records described below; downloaded resources, model weights and runtime records are not included in Git. Split indices are stored in `outputs/main/splits.json`. Per-method directories hold the test predictions, confusion matrices, classification reports, and validation selection records. BERT's three-epoch history is stored in `outputs/main/bert/training_history.json`. Library versions and run arguments are saved alongside the aggregate results.

B.2 Public Resource Locations

The download script uses the following public resources:

- NYT: `https://github.com/yumeng5/WeSHClass/tree/master/nyt`.
- AG News: `https://huggingface.co/datasets/fancyzhx/ag_news`.
- GloVe: `https://nlp.stanford.edu/projects/glove/`.
- BERT: `https://huggingface.co/google-bert/bert-base-uncased`.

GloVe is downloaded through Gensim's `glove-wiki-gigaword-100` mirror of the same Wikipedia 2014 and Gigaword 5 vectors. The script removes the word2vec-format header before loading the vectors.

C Complete Supplementary Experimental Protocol

C.1 Supplementary Experiment Design

The fixed-partition context study evaluates BERT at input lengths 64, 128, 256, and 512 with uniform or inverse-frequency-weighted cross-entropy, each using training seeds 42, 43, and 44. These 24 settings retain three epochs, batch size 32, the same optimizer, and validation-based checkpoint selection. Class k receives weight $w_k = N_{\text{train}}/(29N_k)$, where N_k is its training count. Weighted cross-entropy is normalized by the sum of example weights within each batch.

For a matched-text comparison, each BoW representation receives the original character span covered by BERT's first $L - 2$ WordPiece tokens. Its vocabulary and IDF are fitted on these training prefixes. Embedding pooling is extended to

$$e_{\text{idf}}(d) = \frac{\sum_{w \in K_d} \text{idf}(w) e(w)}{\sum_{w \in K_d} \text{idf}(w)}. \tag{7}$$

Here K_d retains token occurrences, and smoothed IDF uses NYT training document frequencies. A known embedding word absent from training receives document frequency zero. Each Word2Vec source is trained with all three seeds; both poolings use each learned model. GloVe stays fixed. Low-frequency F1 averages classes with at most 100 training examples, including 12 classes in the reference partition.

C.2 Repeated Partitions and Training Budgets

Partition seeds 42–46 each produce a shuffled 80/10/10 split using the same cleaning procedure. Within every partition, BERT crosses lengths 256 and 512, both losses, and three training seeds. All three sparse methods cross full text and matched 256/512-token prefixes with both losses. NYT Word2Vec uses three training seeds, IDF pooling, and both logistic-regression losses. Each partition refits its vocabulary, IDF, scaler, NYT embeddings, class weights, and validation selection. Sparse classifiers are fitted once per setting because the deterministic solver has no training-seed study.

On partition 42, BERT also crosses lengths 64, 256, and 512 with both losses and a ten-epoch budget. The learning-rate schedule spans the specified training horizon, and the best validation checkpoint is selected from all completed epochs. A position study uses weighted 256-token BERT with the first 254 WordPieces, the last 254, or the first and last 127 concatenated. Shorter documents remain intact. These controls use three training seeds.

The repeated-partition package contains 204 configurations, of which 15 reuse identical fixed-partition settings. Together the studies contain 233 distinct configurations. For repeated-partition summaries, we first average training seeds within each partition, then report the mean and sample standard deviation of the five partition means. Fixed-partition error bars instead describe the three training seeds.

D SUPPLEMENTARY CONFIGURATION AND CLASS RESULTS

Table 13 summarizes the pooling comparison. Tables 14–16 give the 41 supplementary configurations, including four references from the main comparison. Every test score is recomputed from saved predictions. Training seeds vary initialization, sampling, and training order while the data indices remain fixed. Class weights and IDF use only the training partition. Per-run records and seed summaries are stored in `outputs/explorations/`.

Table 12: Sparse models on full documents or matched text prefixes in partition 42. Scores are test percentages. Each representation fits its training vocabulary and selects C on validation data.

	Binary BoW		Word Frequency		TF-IDF	
Budget	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1
Full	92.47	80.47	93.63	83.61	94.01	82.85
64	84.79	71.43	84.95	71.31	86.64	73.47
128	87.25	67.59	88.86	73.87	90.78	76.36
256	91.17	73.59	91.01	77.39	93.01	79.54

Table 13: Embedding pooling comparison on NYT. Scores are percentages. Word2Vec results are means $\pm$ sample standard deviations over three paired training seeds; GloVe is a single fixed-vector experiment.

		Mean pooling		TF-IDF weighting	
Embedding	Seeds	Acc.	Macro-F1	Acc.	Macro-F1
GloVe	1	91.71	79.38	91.86	79.92
Word2Vec (AG News)	3	90.19 ± 0.16	77.09 ± 0.94	90.63 ± 0.28	78.45 ± 0.04
Word2Vec (NYT)	3	92.40 ± 0.20	80.65 ± 1.10	92.73 ± 0.59	81.65 ± 0.59

Table 14: Individual BERT runs. F1 columns denote Macro-F1 except Rare F1; Zero-F1 counts classes with zero test F1. All scores are percentages.

Length	Loss	Seed	Epoch	Val. F1	Test Acc.	Test F1	Rare F1	Zero-F1
64	Uniform	42	3	63.37	89.63	62.97	30.55	7
64	Uniform	43	3	65.92	90.48	69.26	44.21	4
64	Uniform	44	3	65.91	89.78	64.57	33.97	6
64	Weighted	42	3	80.57	89.86	80.40	74.17	0
64	Weighted	43	3	78.74	89.71	80.33	74.81	0
64	Weighted	44	3	80.05	89.48	77.84	69.58	0
128	Uniform	42	3	72.60	91.94	69.05	41.59	5
128	Uniform	43	3	71.36	92.55	74.14	54.68	3
128	Uniform	44	3	75.61	91.86	73.95	55.22	3
128	Weighted	42	3	82.49	91.71	82.20	75.02	0
128	Weighted	43	3	84.23	91.78	82.28	76.07	0
128	Weighted	44	3	83.49	91.94	82.70	76.49	0
256	Uniform	42	3	73.07	93.55	74.67	53.49	4
256	Uniform	43	3	76.78	93.86	77.18	59.89	3
256	Uniform	44	3	76.60	93.86	78.02	61.56	3
256	Weighted	42	3	85.09	92.86	83.70	77.71	0
256	Weighted	43	3	86.21	92.93	83.30	77.02	0
256	Weighted	44	3	87.27	93.01	82.77	75.29	0

Table 15: Prefix vocabulary sizes, validation selections, and test scores. F1 denotes Macro-F1; scores are percentages.

Representation	Budget	Vocabulary	C	Val. F1	Test Acc.	Test F1
Binary BoW	64	28,353	10	68.47	84.79	71.43
Binary BoW	128	38,001	10	76.27	87.25	67.59
Binary BoW	256	49,632	1	80.87	91.17	73.59
Word Frequency	64	28,353	10	71.30	84.95	71.31
Word Frequency	128	38,001	10	79.26	88.86	73.87
Word Frequency	256	49,632	1	81.37	91.01	77.39
TF-IDF	64	28,353	10	69.65	86.64	73.47
TF-IDF	128	38,001	10	76.57	90.78	76.36
TF-IDF	256	49,632	10	80.02	93.01	79.54

Table 16: Individual embedding pooling runs. F1 denotes Macro-F1, with scores in percent.

Embedding	Pooling	Seed	C	Val. F1	Test Acc.	Test F1
GloVe	Mean	42	0.1	83.72	91.71	79.38
GloVe	TF-IDF	42	0.1	82.18	91.86	79.92
Word2Vec (AG News)	Mean	42	0.1	76.81	90.02	77.11
Word2Vec (AG News)	TF-IDF	42	0.1	77.89	90.32	78.49
Word2Vec (AG News)	Mean	43	0.1	75.91	90.25	76.14
Word2Vec (AG News)	TF-IDF	43	0.1	77.91	90.86	78.41
Word2Vec (AG News)	Mean	44	0.1	75.04	90.32	78.01
Word2Vec (AG News)	TF-IDF	44	0.1	77.11	90.71	78.45
Word2Vec (NYT)	Mean	42	1	83.33	92.63	81.85
Word2Vec (NYT)	TF-IDF	42	0.1	83.53	92.86	81.84
Word2Vec (NYT)	Mean	43	1	82.51	92.24	80.42
Word2Vec (NYT)	TF-IDF	43	1	82.53	93.24	82.12
Word2Vec (NYT)	Mean	44	1	83.39	92.32	79.68
Word2Vec (NYT)	TF-IDF	44	1	83.24	92.09	80.98

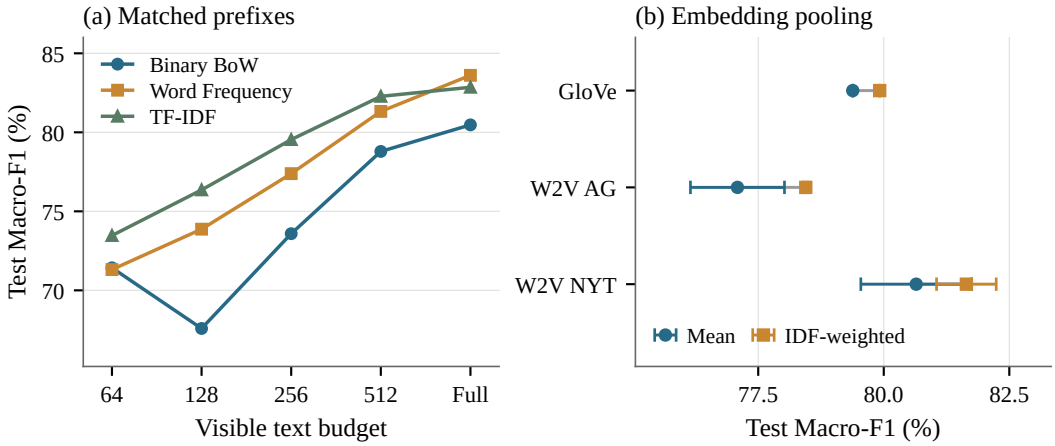


Figure 8: Fixed-partition controls for document coverage and pooling. (a) Sparse-model Macro-F1 on matched prefixes and full documents. (b) Mean and IDF-weighted embedding pooling; Word2Vec error bars are sample standard deviations across three training seeds. GloVe uses fixed vectors.

Table 17: BERT class-level test F1 averaged over three training seeds. Weighted denotes inverse-frequency cross-entropy. Scores are percentages. Class IDs follow Table 8.

		64		128		256	
ID	Class	Uniform	Weighted	Uniform	Weighted	Uniform	Weighted
1	abortion	0.00	100.00	0.00	100.00	0.00	100.00
2	baseball	97.08	97.06	98.49	98.33	99.29	98.98
3	basketball	91.81	91.86	96.47	95.87	98.15	97.87
4	business	77.24	71.04	74.84	70.68	82.74	75.74
5	cosmos	0.00	43.25	0.00	39.18	0.00	39.18
6	dance	90.33	90.67	100.00	100.00	96.00	98.77
7	economy	75.64	77.37	77.79	79.31	81.14	79.98
8	energy companies	57.54	55.34	56.44	57.56	56.79	55.28
9	environment	0.00	62.42	0.00	62.22	0.00	55.00
10	federal budget	0.00	47.37	44.44	67.94	38.18	54.05
11	football	92.98	92.85	96.67	96.44	98.44	98.22
12	gay rights	66.67	83.08	68.69	71.67	73.50	89.74
13	golf	97.52	96.78	98.69	98.54	99.56	99.56
14	gun control	20.00	71.43	73.42	80.00	87.18	80.24
15	hockey	99.33	98.23	99.34	99.34	100.00	100.00
16	immigration	6.67	75.05	46.55	80.00	69.64	86.11
17	international business	80.36	75.00	78.98	74.12	78.32	76.45
18	law enforcement	71.53	74.50	77.38	79.11	84.77	83.57
19	military	64.44	85.71	85.71	85.71	100.00	75.79
20	movies	90.82	90.82	92.82	93.33	93.41	94.74
21	music	93.77	92.74	98.60	99.06	97.68	97.17
22	politics	79.00	80.32	82.10	82.58	85.61	81.45
23	science	82.20	76.02	82.16	73.89	83.40	75.35
24	soccer	98.59	98.58	98.58	98.35	98.58	98.35
25	stocks and bonds	80.83	75.15	78.73	70.62	77.89	76.19
26	surveillance	43.33	82.22	43.33	88.57	86.67	95.24
27	television	83.12	64.96	84.24	83.37	84.42	81.82
28	tennis	98.93	98.51	99.57	99.14	100.00	99.78
29	the affordable care act	62.62	57.90	65.08	64.37	70.69	69.79

E Repeated-Partition and Budget Results

Table 18: Repeated-partition results in percent. Each entry is the mean ± sample standard deviation of five partition means. BERT and NYT Word2Vec average three training seeds within each partition; sparse models use one fit. Classical models receive full documents; BERT receives the stated prefix budget.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	94.18 ± 1.13	84.49 ± 2.75	76.08 ± 4.02
BERT-512 (weighted)	93.62 ± 0.47	84.22 ± 0.87	76.56 ± 1.44
Binary (weighted)	93.87 ± 0.82	83.55 ± 2.26	75.01 ± 3.39
BERT-256 (weighted)	93.29 ± 0.40	83.45 ± 1.27	74.87 ± 2.01
Count (weighted)	93.63 ± 0.88	83.35 ± 1.74	75.02 ± 2.39
TF-IDF (uniform)	94.58 ± 0.82	83.22 ± 2.09	72.32 ± 3.41
Count (uniform)	93.55 ± 0.81	81.68 ± 2.17	71.18 ± 3.87
NYT W2V-IDF (uniform)	93.02 ± 0.49	81.07 ± 0.62	70.51 ± 2.83
Binary (uniform)	93.56 ± 0.96	80.95 ± 2.01	69.02 ± 3.10
NYT W2V-IDF (weighted)	92.09 ± 0.43	80.43 ± 1.37	70.14 ± 2.88
BERT-512 (uniform)	94.54 ± 0.60	80.11 ± 1.34	65.17 ± 2.45
BERT-256 (uniform)	94.14 ± 0.54	77.63 ± 2.10	58.84 ± 4.11

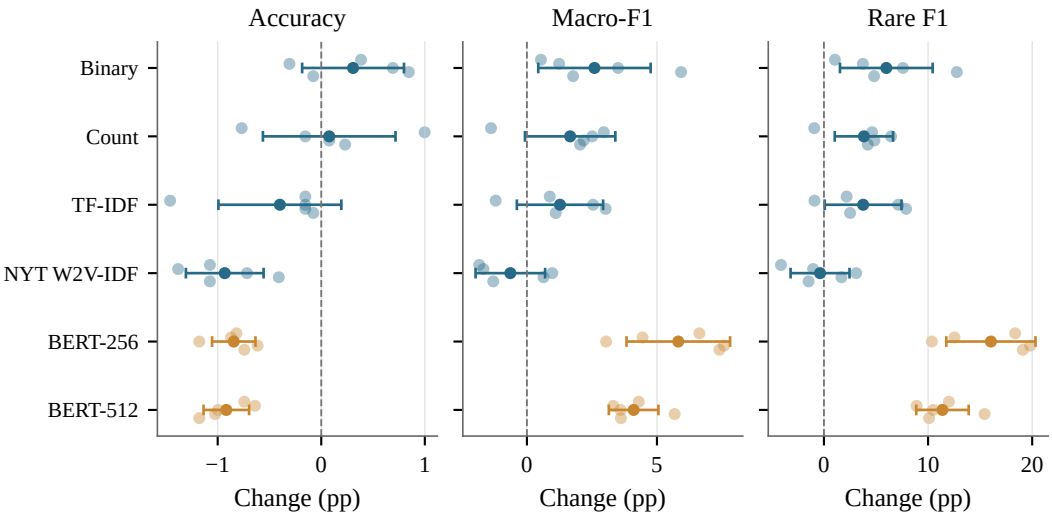


Figure 9: Weighted minus ordinary loss on five paired partitions. Points are partition-level differences after averaging training seeds where applicable. Error bars are the mean ± sample standard deviation of those five differences. All classical representations use full documents.

The complete 204-configuration matrix and all predictions are stored in `outputs/scaling/`. Tables 20–24 expose the individual partition means. Sparse entries are single fits; BERT and NYT Word2Vec entries average three training seeds, with sample standard deviations over those seeds.

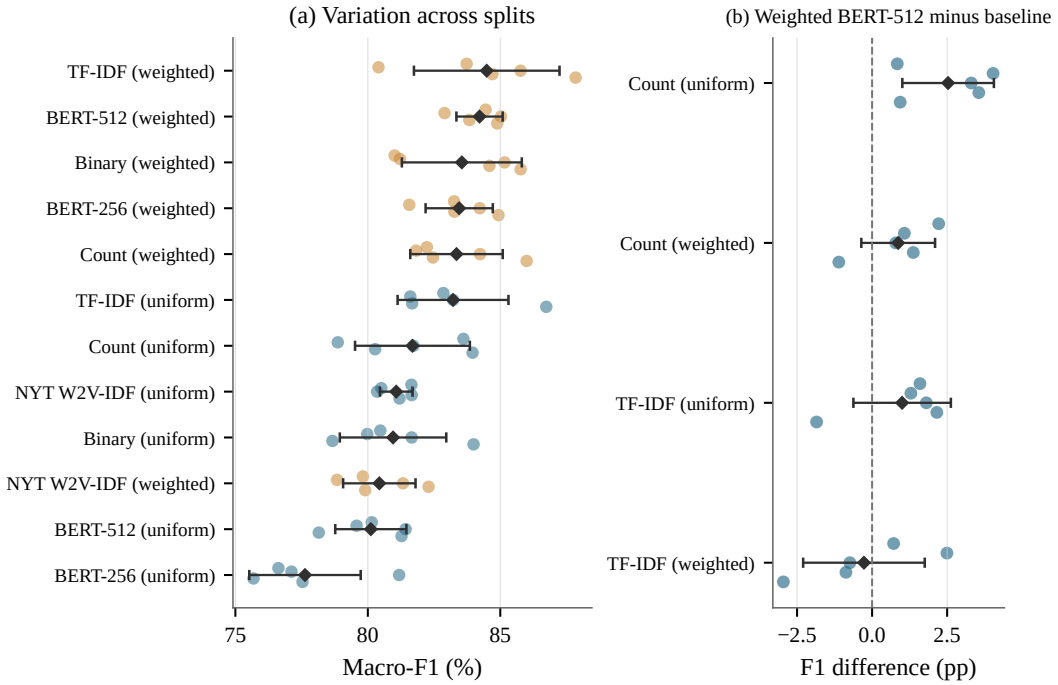


Figure 10: Partition sensitivity. (a) Dots are the five partition means, with a diamond and error bar showing their mean and sample standard deviation. Orange denotes weighted loss; blue denotes ordinary loss. (b) Paired partition-level Macro-F1 differences between weighted BERT-512 and full-document sparse baselines. Error bars describe partition variation.

Table 19: Effect of inverse-frequency weighting. Entries are the mean ± sample standard deviation of five paired partition differences, in percentage points. Training seeds are averaged within each partition.

Method	ΔAccuracy	ΔMacro-F1	ΔRare F1
Binary	0.31 ± 0.49	2.60 ± 2.16	5.99 ± 4.45
Count	0.08 ± 0.64	1.66 ± 1.74	3.84 ± 2.80
TF-IDF	-0.40 ± 0.59	1.27 ± 1.66	3.76 ± 3.68
NYT W2V-IDF	-0.93 ± 0.38	-0.64 ± 1.34	-0.37 ± 2.83
BERT-256	-0.84 ± 0.21	5.82 ± 1.99	16.03 ± 4.29
BERT-512	-0.92 ± 0.22	4.10 ± 0.95	11.38 ± 2.52

Table 20: Partition seed 42. Scores are percentages; error terms measure training-seed variation.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	93.86	83.73	75.89
BERT-512 (weighted)	93.45 ± 0.12	84.45 ± 0.59	78.43 ± 1.42
Binary (weighted)	92.86	81.01	71.56
BERT-256 (weighted)	92.93 ± 0.08	83.26 ± 0.47	76.67 ± 1.25
Count (weighted)	92.86	82.23	75.54
TF-IDF (uniform)	94.01	82.85	73.71
Count (uniform)	93.63	83.61	76.48
NYT W2V-IDF (uniform)	92.73 ± 0.59	81.65 ± 0.59	74.18 ± 0.63
Binary (uniform)	92.47	80.47	70.51
NYT W2V-IDF (weighted)	91.65 ± 0.19	79.81 ± 0.73	70.05 ± 1.58
BERT-512 (uniform)	94.19 ± 0.23	80.15 ± 1.74	66.43 ± 4.73
BERT-256 (uniform)	93.75 ± 0.18	76.62 ± 1.74	58.31 ± 4.26

Table 21: Partition seed 43, with the same reporting convention as Table 20.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	92.78	80.40	69.47
BERT-512 (weighted)	93.42 ± 0.42	82.90 ± 0.55	75.03 ± 0.31
Binary (weighted)	93.39	81.21	71.07
BERT-256 (weighted)	93.11 ± 1.17	81.56 ± 3.58	71.48 ± 6.60
Count (weighted)	93.86	81.82	71.36
TF-IDF (uniform)	94.24	81.60	70.38
Count (uniform)	92.86	78.87	66.74
NYT W2V-IDF (uniform)	93.11 ± 0.23	80.51 ± 1.18	69.19 ± 2.54
Binary (uniform)	93.70	79.98	67.33
NYT W2V-IDF (weighted)	91.73 ± 0.23	78.84 ± 0.83	68.14 ± 1.92
BERT-512 (uniform)	94.06 ± 0.35	79.57 ± 2.45	66.12 ± 4.88
BERT-256 (uniform)	93.98 ± 0.12	77.12 ± 3.09	58.95 ± 7.76

Table 22: Partition seed 44. Scores are percentages; error terms measure training-seed variation.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	95.01	85.76	77.22
BERT-512 (weighted)	94.09 ± 0.13	85.02 ± 1.14	77.57 ± 3.17
Binary (weighted)	94.55	85.16	76.99
BERT-256 (weighted)	93.68 ± 0.51	84.23 ± 0.67	76.02 ± 0.66
Count (weighted)	93.70	84.24	75.75
TF-IDF (uniform)	95.16	83.22	70.08
Count (uniform)	93.86	81.72	69.30
NYT W2V-IDF (uniform)	93.16 ± 0.31	80.35 ± 0.65	67.77 ± 0.91
Binary (uniform)	93.86	81.65	69.41
NYT W2V-IDF (weighted)	92.45 ± 0.23	81.33 ± 0.34	70.87 ± 1.00
BERT-512 (uniform)	95.08 ± 0.20	81.42 ± 2.12	67.10 ± 5.06
BERT-256 (uniform)	94.85 ± 0.58	81.18 ± 2.58	65.65 ± 5.50

Table 23: Partition seed 45, with the same reporting convention as Table 20.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	93.63	84.70	77.61
BERT-512 (weighted)	93.04 ± 0.22	83.83 ± 0.22	76.37 ± 0.38
Binary (weighted)	93.70	84.59	77.61
BERT-256 (weighted)	92.96 ± 0.36	83.27 ± 0.32	75.02 ± 0.95
Count (weighted)	92.78	82.46	74.53
TF-IDF (uniform)	93.78	81.67	69.72
Count (uniform)	92.70	80.27	69.69
NYT W2V-IDF (uniform)	92.40 ± 0.31	81.66 ± 1.29	72.86 ± 2.30
Binary (uniform)	92.86	78.66	64.85
NYT W2V-IDF (weighted)	91.99 ± 0.51	82.29 ± 1.00	74.54 ± 2.24
BERT-512 (uniform)	94.06 ± 0.16	78.15 ± 0.85	60.94 ± 2.26
BERT-256 (uniform)	93.57 ± 0.12	75.68 ± 0.83	55.21 ± 2.54

Table 24: Partition seed 46. Scores are percentages; error terms measure training-seed variation.

Method and loss	Accuracy	Macro-F1	Rare F1
TF-IDF (weighted)	95.62	87.84	80.23
BERT-512 (weighted)	94.11 ± 0.25	84.89 ± 0.43	75.38 ± 1.10
Binary (weighted)	94.85	85.77	77.81
BERT-256 (weighted)	93.78 ± 0.48	84.93 ± 0.57	75.17 ± 1.00
Count (weighted)	94.93	86.00	77.93
TF-IDF (uniform)	95.70	86.74	77.72
Count (uniform)	94.70	83.95	73.72
NYT W2V-IDF (uniform)	93.70 ± 0.08	81.20 ± 0.25	68.57 ± 0.95
Binary (uniform)	94.93	83.99	73.00
NYT W2V-IDF (weighted)	92.63 ± 0.15	79.90 ± 0.70	67.09 ± 2.15
BERT-512 (uniform)	95.29 ± 0.09	81.27 ± 1.61	65.28 ± 3.91
BERT-256 (uniform)	94.52 ± 0.12	77.53 ± 1.51	56.07 ± 3.88

Table 25: Sparse models with matched 256/512-token prefixes or full documents. Entries are the mean $\pm$ sample standard deviation of five partition scores. All weights, vocabularies and IDF values are fitted on the corresponding training partition.

Text budget	Method and loss	Accuracy	Macro-F1	Rare F1
Full	Binary (uniform)	93.56 ± 0.96	80.95 ± 2.01	69.02 ± 3.10
Full	Binary (weighted)	93.87 ± 0.82	83.55 ± 2.26	75.01 ± 3.39
Full	Count (uniform)	93.55 ± 0.81	81.68 ± 2.17	71.18 ± 3.87
Full	Count (weighted)	93.63 ± 0.88	83.35 ± 1.74	75.02 ± 2.39
Full	TF-IDF (uniform)	94.58 ± 0.82	83.22 ± 2.09	72.32 ± 3.41
Full	TF-IDF (weighted)	94.18 ± 1.13	84.49 ± 2.75	76.08 ± 4.02
256	Binary (uniform)	91.61 ± 0.68	76.79 ± 3.37	63.15 ± 7.38
256	Binary (weighted)	92.09 ± 0.83	80.21 ± 2.11	70.32 ± 4.02
256	Count (uniform)	91.89 ± 0.74	78.96 ± 2.05	68.43 ± 4.50
256	Count (weighted)	91.84 ± 0.62	79.67 ± 1.43	70.05 ± 3.46
256	TF-IDF (uniform)	92.60 ± 0.68	78.27 ± 2.26	64.76 ± 4.94
256	TF-IDF (weighted)	92.06 ± 1.10	81.86 ± 2.08	73.96 ± 4.03
512	Binary (uniform)	93.41 ± 1.08	80.03 ± 2.72	66.86 ± 4.65
512	Binary (weighted)	93.29 ± 1.14	81.67 ± 2.53	71.00 ± 3.04
512	Count (uniform)	92.84 ± 0.58	80.65 ± 1.94	70.24 ± 3.59
512	Count (weighted)	93.03 ± 0.42	82.08 ± 2.16	73.24 ± 4.45
512	TF-IDF (uniform)	93.92 ± 0.72	82.09 ± 2.11	70.93 ± 4.59
512	TF-IDF (weighted)	93.63 ± 1.04	83.72 ± 2.63	75.62 ± 4.53

E.1 Optimization Budget

The budget comparison retains the same inputs, initialization seeds, optimizer constants and checkpoint criterion. Each run completes its specified horizon, including the matching warmup and decay schedule. Selected epochs in Table 26 correspond to training seeds 42, 43 and 44, in that order.

Table 26: Three- versus ten-epoch BERT budgets on partition 42. Scores are means ± sample standard deviations over three training seeds. Selected lists give the validation-selected epochs.

Length	Loss	Epochs	Accuracy	Macro-F1	Rare F1	Selected
64	Uniform	3	89.96 ± 0.45	65.60 ± 3.27	36.24 ± 7.11	3,3,3
64	Uniform	10	90.58 ± 0.51	79.80 ± 1.96	72.81 ± 3.31	10,10,8
64	Weighted	3	89.68 ± 0.19	79.53 ± 1.46	72.85 ± 2.85	3,3,3
64	Weighted	10	90.63 ± 0.46	80.59 ± 0.37	74.25 ± 0.95	4,10,6
256	Uniform	3	93.75 ± 0.18	76.62 ± 1.74	58.31 ± 4.26	3,3,3
256	Uniform	10	94.29 ± 0.66	85.48 ± 2.84	78.49 ± 5.58	6,9,4
256	Weighted	3	92.93 ± 0.08	83.26 ± 0.47	76.67 ± 1.25	3,3,3
256	Weighted	10	93.75 ± 0.29	84.46 ± 0.81	77.41 ± 0.83	4,4,4
512	Uniform	3	94.19 ± 0.23	80.15 ± 1.74	66.43 ± 4.73	3,3,3
512	Uniform	10	94.96 ± 0.29	86.38 ± 0.51	79.53 ± 0.91	5,10,4
512	Weighted	3	93.45 ± 0.12	84.45 ± 0.59	78.43 ± 1.42	3,3,3
512	Weighted	10	94.21 ± 0.70	85.72 ± 1.33	79.58 ± 1.83	9,3,4

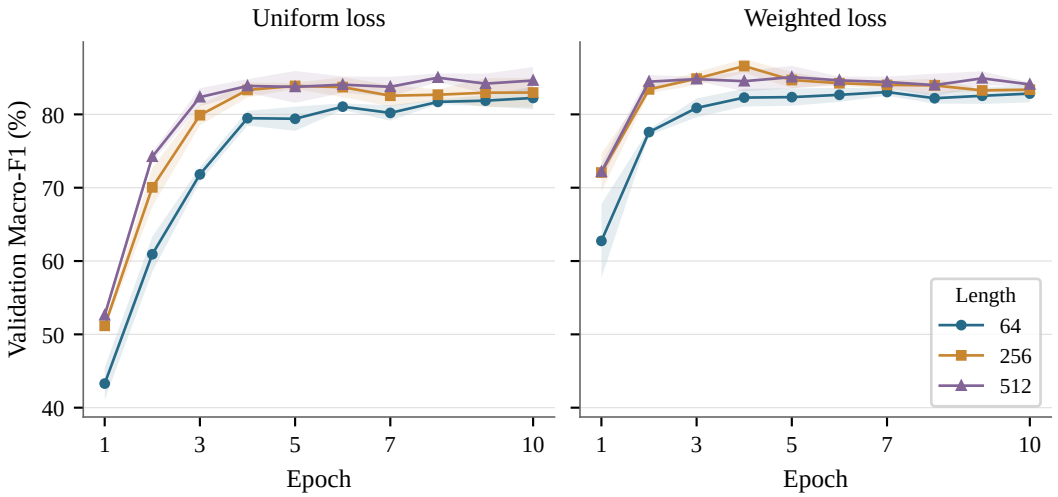


Figure 11: Validation Macro-F1 throughout ten-epoch training at three context lengths. Lines are three-seed means and shaded bands are sample standard deviations. Checkpoints are selected separately for each run.

E.2 Text Position and Class-Level Weighting

Table 27: Text-position control for weighted BERT-256 on partition 42. Every run uses three epochs and 254 text-token positions. Head + tail concatenates the first and last 127 WordPieces. Scores are three-seed means ± sample standard deviations.

Text position	Accuracy	Macro-F1	Rare F1
Head	92.93 ± 0.08	83.26 ± 0.47	76.67 ± 1.25
Tail	89.99 ± 0.25	76.22 ± 0.37	63.70 ± 1.59
Head + tail	92.45 ± 0.09	81.57 ± 1.31	72.51 ± 3.01

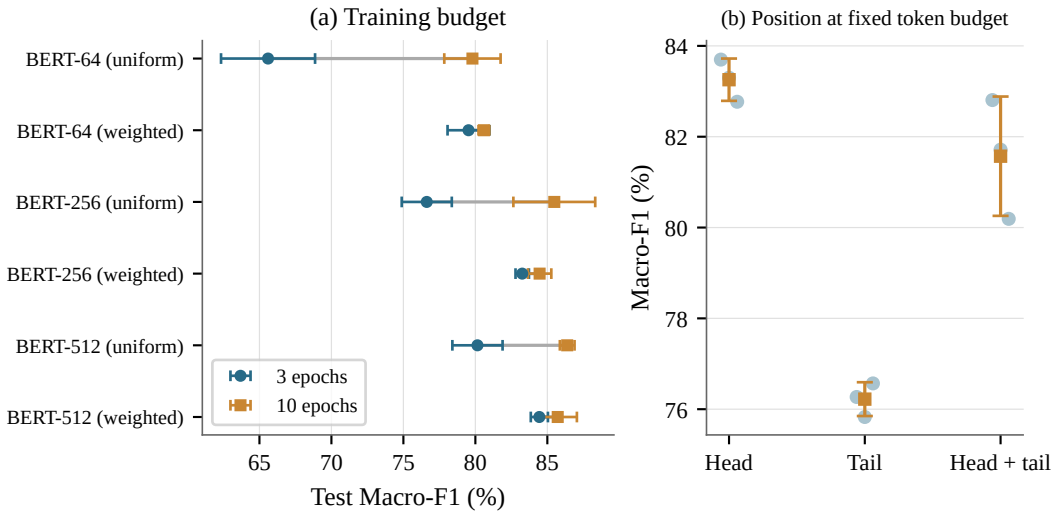


Figure 12: Fixed-partition controls. (a) Test Macro-F1 under three- and ten-epoch budgets, with sample standard deviations over three seeds. (b) The location of retained tokens at a fixed weighted BERT-256 budget; faint dots are individual seeds.

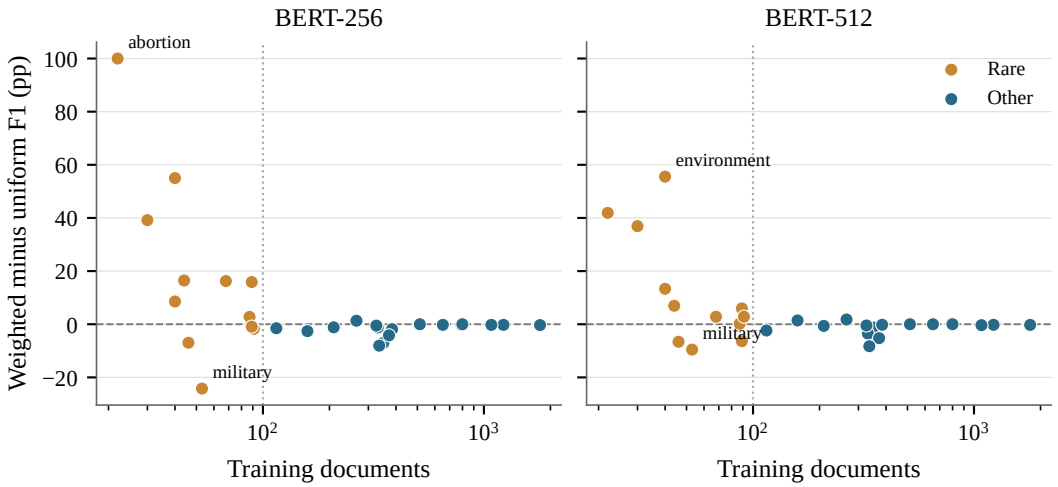


Figure 13: Training frequency and the effect of weighting on partition 42. Each point is a category's mean weighted-minus-ordinary F1 across three paired seeds. The horizontal axis is logarithmic; orange marks categories with at most 100 training documents. Negative values indicate lower F1 under weighting.